Nigeria's Wood Trade Report

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0.1 Introduction

The Food and Agriculture Organization (FAO) tracks forestry production and trade data globally, providing vital statistics for understanding economic and environmental trends. For Nigeria, the wood and paper-product sector plays a significant role in the national economy, yet understanding its trade dynamics requires careful data management and exploration.

This report presents a comprehensive, reproducible analysis of Nigeria's wood and paper-product trade data from 2000 to 2025 (Food and Agriculture Organization, 2025). The objective is to build a clean, documented dataset and conduct an exploratory data analysis (EDA) to uncover useful patterns that can inform policy decisions.

The report is structured into two main sections:

I. Data Management & Wrangling: We begin by importing the raw FAOSTAT data, cleaning the structure, standardizing variable names, and filtering out non-relevant metrics to ensure mathematical validity. We then feature engineer the data to convert raw units into standardized million USD values and isolate leaf-level products from hierarchical aggregates to prevent double-counting.

II. Exploratory Data Analysis (EDA): Using the cleaned data, we visualize and compare imports and exports over time. We investigate long-term trends in total versus average trade values, track the expansion of product diversity, and identify the top export products.

By providing a clean dataset alongside clear visualizations, this report aims to provide the policy team with an evidence-based understanding of Nigeria's trade footprint in the forestry sector.

0.2 Part 1: Data Management

1. Load Packages

To ensure the analysis is reproducible, I used a package-management approach by loading all required libraries at the beginning. `tidyverse` handles the data wrangling and visualization, `janitor` cleans the column names, and `skimr` is used for data inspection.

```
library(tidyverse)
library(janitor)
library(readr)
library(skimr)
library(dplyr)
library(car)
```

2. Import Data and Clean Column Names

The raw CSV file that contains the data were imported directly from the GitHub repository. The function `janitor::clean_names()` was used to convert all column names into a standard snake_case convention, which makes them easier to reference in R.

```
nigeria_wood_trade_raw <- read_csv("https://raw.githubusercontent.com/EU-Study-Assist/data-f
  janitor::clean_names()
```

```
Rows: 9110 Columns: 15
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (8): Domain Code, Domain, Area, Element, Item, Unit, Flag, Flag Description
dbl (6): Area Code (M49), Element Code, Item Code, Year Code, Year, Value
lgl (1): Note

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(nigeria_wood_trade_raw)
```

```
# A tibble: 6 x 15
  domain_code domain    area_code_m49 area  element_code element item_code item
  <chr>       <chr>             <dbl> <chr>        <dbl> <chr>       <dbl> <chr>
1 FO          Forestry~           566 Nige~         5622 Import~      1659 Wood~
2 FO          Forestry~           566 Nige~         5922 Export~      1659 Wood~
3 FO          Forestry~           566 Nige~         5622 Import~      1659 Wood~
```

```
4 FO          Forestry~           566 Nige~         5922 Export~      1659 Wood~
5 FO          Forestry~           566 Nige~         5622 Import~      1659 Wood~
6 FO          Forestry~           566 Nige~         5922 Export~      1659 Wood~
# i 7 more variables: year_code <dbl>, year <dbl>, unit <chr>, value <dbl>,
#   flag <chr>, flag_description <chr>, note <lgl>
```

```
nigeria_wood_trade_new <- nigeria_wood_trade_raw |>
  clean_names()

nigeria_wood_trade_new
```

```
# A tibble: 9,110 x 15
   domain_code domain   area_code_m49 area  element_code element item_code item
   <chr>       <chr>            <dbl> <chr>        <dbl> <chr>       <dbl> <chr>
 1 FO          Forestr~           566 Nige~         5622 Import~      1659 Wood~
 2 FO          Forestr~           566 Nige~         5922 Export~      1659 Wood~
 3 FO          Forestr~           566 Nige~         5622 Import~      1659 Wood~
 4 FO          Forestr~           566 Nige~         5922 Export~      1659 Wood~
 5 FO          Forestr~           566 Nige~         5622 Import~      1659 Wood~
 6 FO          Forestr~           566 Nige~         5922 Export~      1659 Wood~
 7 FO          Forestr~           566 Nige~         5622 Import~      1659 Wood~
 8 FO          Forestr~           566 Nige~         5922 Export~      1659 Wood~
 9 FO          Forestr~           566 Nige~         5622 Import~      1659 Wood~
10 FO          Forestr~           566 Nige~         5922 Export~      1659 Wood~
# i 9,100 more rows
# i 7 more variables: year_code <dbl>, year <dbl>, unit <chr>, value <dbl>,
#   flag <chr>, flag_description <chr>, note <lgl>
```

Note: The snake_case conversion was saved using the name `nigeria_wood_trade_new`

3. Inspect the Imported Object Using `skim()` and `str()`, Dataset were inspected to know the structure of the data. The function `head()`, `tail()` and `car::some` were used to preview the dataset to understand it more

```
# Structure and summary
str(nigeria_wood_trade_raw)
```

```
spc_tbl_ [9,110 x 15] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ domain_code     : chr [1:9110] "FO" "FO" "FO" "FO" ...
 $ domain          : chr [1:9110] "Forestry Production and Trade" "Forestry Production and Tr
 $ area_code_m49   : num [1:9110] 566 566 566 566 566 566 566 566 566 566 ...
```

```
$ area            : chr [1:9110] "Nigeria" "Nigeria" "Nigeria" "Nigeria" ...
$ element_code    : num [1:9110] 5622 5922 5622 5922 5622 ...
$ element         : chr [1:9110] "Import value" "Export value" "Import value" "Export value"
$ item_code       : num [1:9110] 1659 1659 1659 1659 1659 ...
$ item            : chr [1:9110] "Wood and paper products (export/import)" "Wood and paper p
$ year_code       : num [1:9110] 2000 2000 2001 2001 2002 ...
$ year            : num [1:9110] 2000 2000 2001 2001 2002 ...
$ unit            : chr [1:9110] "1000 USD" "1000 USD" "1000 USD" "1000 USD" ...
$ value           : num [1:9110] 181796 52276 170423 36587 174679 ...
$ flag            : chr [1:9110] "A" "E" "A" "E" ...
$ flag_description: chr [1:9110] "Official figure" "Estimated value" "Official figure" "Esti
$ note            : logi [1:9110] NA NA NA NA NA NA ...
- attr(*, "spec")=
 .. cols(
 ..   `Domain Code` = col_character(),
 ..   Domain = col_character(),
 ..   `Area Code (M49)` = col_double(),
 ..   Area = col_character(),
 ..   `Element Code` = col_double(),
 ..   Element = col_character(),
 ..   `Item Code` = col_double(),
 ..   Item = col_character(),
 ..   `Year Code` = col_double(),
 ..   Year = col_double(),
 ..   Unit = col_character(),
 ..   Value = col_double(),
 ..   Flag = col_character(),
 ..   `Flag Description` = col_character(),
 ..   Note = col_logical()
 .. )
- attr(*, "problems")=<pointer: 0x0000018279bffc80>
```

```
skim_without_charts(nigeria_wood_trade_raw)
```

Table 1: Data summary

Name	nigeria_wood_trade_raw
Number of rows	9110
Number of columns	15
Column type frequency:	
character	8

logical	1
numeric	6
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
domain_code	0	1	2	2	0	1	0
domain	0	1	29	29	0	1	0
area	0	1	7	7	0	1	0
element	0	1	10	15	0	5	0
item	0	1	8	69	0	80	0
unit	0	1	1	8	0	3	0
flag	0	1	1	1	0	4	0
flag_description	0	1	15	35	0	4	0

Variable type: logical

skim_variable	n_missing	complete_rate	mean	count
note	9110	0	NaN	:

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
area_code_m49	0	1	566.00	0.00	566	566	566	566	566
element_code	0	1	5701.45	161.40	5510	5610	5622	5916	5922
item_code	0	1	1763.82	285.18	1603	1640	1674	1868	5225
year_code	0	1	2001.40	16.89	1961	1990	2005	2015	2024
year	0	1	2001.40	16.89	1961	1990	2005	2015	2024
value	0	1	1860831.21	9574805.49	0	112	2600	35280	78399506

```
# Preview the dataset (top, bottom, and a random sample)
head(nigeria_wood_trade_raw)
```

```
# A tibble: 6 x 15
  domain_code domain    area_code_m49 area  element_code element item_code item
  <chr>       <chr>             <dbl> <chr>        <dbl> <chr>       <dbl> <chr>
1 FO          Forestry~           566 Nige~         5622 Import~      1659 Wood~
2 FO          Forestry~           566 Nige~         5922 Export~      1659 Wood~
3 FO          Forestry~           566 Nige~         5622 Import~      1659 Wood~
4 FO          Forestry~           566 Nige~         5922 Export~      1659 Wood~
5 FO          Forestry~           566 Nige~         5622 Import~      1659 Wood~
6 FO          Forestry~           566 Nige~         5922 Export~      1659 Wood~
# i 7 more variables: year_code <dbl>, year <dbl>, unit <chr>, value <dbl>,
#   flag <chr>, flag_description <chr>, note <lgl>
```

```
tail(nigeria_wood_trade_raw)
```

```
# A tibble: 6 x 15
  domain_code domain    area_code_m49 area  element_code element item_code item
  <chr>       <chr>             <dbl> <chr>        <dbl> <chr>       <dbl> <chr>
1 FO          Forestry~           566 Nige~         5622 Import~      1678 Othe~
2 FO          Forestry~           566 Nige~         5922 Export~      1678 Othe~
3 FO          Forestry~           566 Nige~         5622 Import~      1678 Othe~
4 FO          Forestry~           566 Nige~         5922 Export~      1678 Othe~
5 FO          Forestry~           566 Nige~         5622 Import~      1678 Othe~
6 FO          Forestry~           566 Nige~         5922 Export~      1678 Othe~
# i 7 more variables: year_code <dbl>, year <dbl>, unit <chr>, value <dbl>,
#   flag <chr>, flag_description <chr>, note <lgl>
```

```
# Random sample of rows to see the variety of data
set.seed(3423)
car::some(nigeria_wood_trade_raw)
```

```
# A tibble: 10 x 15
   domain_code domain   area_code_m49 area  element_code element item_code item
   <chr>       <chr>            <dbl> <chr>        <dbl> <chr>       <dbl> <chr>
 1 FO          Forestr~           566 Nige~         5922 Export~      1877 Prim~
 2 FO          Forestr~           566 Nige~         5516 Produc~      1862 Roun~
 3 FO          Forestr~           566 Nige~         5916 Export~      1866 Indu~
 4 FO          Forestr~           566 Nige~         5916 Export~      1670 Indu~
 5 FO          Forestr~           566 Nige~         5916 Export~      1873 Wood~
 6 FO          Forestr~           566 Nige~         5916 Export~      1874 Fibr~
 7 FO          Forestr~           566 Nige~         5622 Import~      1668 Pulp~
 8 FO          Forestr~           566 Nige~         5622 Import~      1669 Reco~
```

```
 9 FO          Forestr~           566 Nige~         5610 Import~      1622 Othe~
10 FO          Forestr~           566 Nige~         5622 Import~      1691 Wood~
# i 7 more variables: year_code <dbl>, year <dbl>, unit <chr>, value <dbl>,
#   flag <chr>, flag_description <chr>, note <lgl>
```

Observation: The dataset contains 9110 rows and 15 columns. Some columns like domain, area, and element_code have no variance (they are constant across all rows). The *year* column was initially read as a character type instead of numeric, which will need to be fixed later. Using `car::some()` makes it clear that the dataset contains a mix of financial values (in *1000 USD*) and physical volumes (in t or m3), which means it must be filter by unit for further analysis.

4. Remove Unneeded Columns and Duplicates

Columns were removed that provided no analytical value for this specific project, such as *codes*, *labels*, and *metadata* (e.g., *domain_code*, *area_code_m49*, *item_code*). Duplicate rows were checked using a combination of *element*, *item*, and *year*. (*Note:* Dataset were filtered strictly between 2000 and 2025 to match the project scope).

```
nigeria_wood_trade_clean <- nigeria_wood_trade_new |>
  select(
    -domain_code,
    -domain,
    -area_code_m49,
    -area,
    -element_code,
    -item_code,
    -year_code,
    -note
  ) |>
  filter(between(year, 2000, 2025))

nigeria_wood_trade_clean
```

```
# A tibble: 5,616 x 7
   element      item                    year unit   value flag  flag_description
   <chr>        <chr>                  <dbl> <chr>  <dbl> <chr> <chr>
 1 Import value Wood and paper produc~  2000 1000~ 181796 A     Official figure
 2 Export value Wood and paper produc~  2000 1000~  52276 E     Estimated value
 3 Import value Wood and paper produc~  2001 1000~ 170423 A     Official figure
 4 Export value Wood and paper produc~  2001 1000~  36587 E     Estimated value
 5 Import value Wood and paper produc~  2002 1000~ 174679 E     Estimated value
 6 Export value Wood and paper produc~  2002 1000~  40494 E     Estimated value
```

```
 7 Import value Wood and paper produc~  2003 1000~ 233876 E     Estimated value
 8 Export value Wood and paper produc~  2003 1000~  32652 E     Estimated value
 9 Import value Wood and paper produc~  2004 1000~ 320050 E     Estimated value
10 Export value Wood and paper produc~  2004 1000~  52546 E     Estimated value
# i 5,606 more rows
```

```r
# Check for duplicates
n_dupes <- sum(duplicated(nigeria_wood_trade_clean[, c("element", "item", "year")]))

n_dupes
```

```
[1] 0
```

Explanation: 0 duplicated rows were found. Therefore, all rows in the dataset are unique and valid, meaning we do not need to remove any data at this stage.

5. Export and Re-import the Cleaned Dataset

Created a *processed_data* folder and exported the cleaned dataset as a CSV file. To demonstrate that the file is readable, it was imported back into R in a separate code block.

```r
# Create folder and export
dir.create("processed_data", showWarnings = FALSE)

write_csv(nigeria_wood_trade_clean, "processed_data/nigeria_wood_trade_clean.csv")

# Re-import to demonstrate readability
reimported_data <- read_csv("processed_data/nigeria_wood_trade_clean.csv")
```

```
Rows: 5616 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (5): element, item, unit, flag, flag_description
dbl (2): year, value

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(reimported_data)
```

```
# A tibble: 6 x 7
  element      item                      year unit   value flag  flag_description
  <chr>        <chr>                    <dbl> <chr>  <dbl> <chr> <chr>
1 Import value Wood and paper product~  2000 1000~ 181796 A     Official figure
2 Export value Wood and paper product~  2000 1000~  52276 E     Estimated value
3 Import value Wood and paper product~  2001 1000~ 170423 A     Official figure
4 Export value Wood and paper product~  2001 1000~  36587 E     Estimated value
5 Import value Wood and paper product~  2002 1000~ 174679 E     Estimated value
6 Export value Wood and paper product~  2002 1000~  40494 E     Estimated value
```

0.3 Part 2: Data Wrangling

1. Reshape the Values of Element

The assignment asks to reshape the values of element into new columns. However, for this specific analysis, reshaping the data is unnecessary because it have no impact on the analysis. Reshaping the data is not necessary because we need to keep the **unit** column separate to avoid mixing USD, tons (t), and cubic meters (m3). Furthermore, keeping the data in its "long" format makes it much easier to use `group_by()` for summary statistics (like calculating total and average values) across different trade directions. However, the crucial operation that must be performed before reshaping is filtering by unit (unit == "1000 USD"). This is highlighted below;

```
# Applying Filter first
element_col <- nigeria_wood_trade_clean |>
  filter(
    element %in% c("Import value", "Export value"),
    unit == "1000 USD"
  )

# Reshaping

element_col |>
  select(year, item, element, value) |>
  pivot_wider(
    names_from = element,
    values_from = value
  )
```

```
# A tibble: 1,569 x 4
    year item                                    `Import value` `Export value`
   <dbl> <chr>                                            <dbl>          <dbl>
```

```
 1  2000 Wood and paper products (export/import)         181796          52276
 2  2001 Wood and paper products (export/import)         170423          36587
 3  2002 Wood and paper products (export/import)         174679          40494
 4  2003 Wood and paper products (export/import)         233876          32652
 5  2004 Wood and paper products (export/import)         320050          52546
 6  2005 Wood and paper products (export/import)         295802          54880
 7  2006 Wood and paper products (export/import)         307959          54160
 8  2007 Wood and paper products (export/import)         400033          94634
 9  2008 Wood and paper products (export/import)         565721          73493
10  2009 Wood and paper products (export/import)         702153         101684
# i 1,559 more rows
```

Note: If you reshape the data first, R will try to create columns for every single unit (e.g., "Import value_1000 USD", "Import value_m3"), creating a massive, fragmented table. By filtering first, we ensure we are only pivoting the financial values.

2. Replacing NA with Zero `(values_fill = 0)`

The argument `values_fill = 0` should not be added. If missing `(NA)` values are replaced with zero, it implies that a specific product had exactly zero trade in that year. This would artificially drag down average value calculations and distort the time-series charts. It is safer to leave NAs as they are and handle them explicitly using `na.rm = TRUE` inside summary functions.

3 & 4. Convert Data Types and Year

Ensuring the columns have the correct data types is essential. The year column should be an integer (not a double). Years are discrete, whole numbers used as labels for time; treating them as continuous decimals (doubles) consumes unnecessary memory and can cause plotting errors on the x-axis.

```r
nigeria_wood_trade_clean <- nigeria_wood_trade_clean |>
  mutate(
    year = as.integer(year),
    value = as.double(value)
  )

head(nigeria_wood_trade_clean, 20) |>
  knitr::kable()
```

element	item	year	unit	value	flag	flag_description
Import value	Wood and paper products (export/import)	2000	1000 USD	181796	A	Official figure
Export value	Wood and paper products (export/import)	2000	1000 USD	52276	E	Estimated value
Import value	Wood and paper products (export/import)	2001	1000 USD	170423	A	Official figure
Export value	Wood and paper products (export/import)	2001	1000 USD	36587	E	Estimated value
Import value	Wood and paper products (export/import)	2002	1000 USD	174679	E	Estimated value
Export value	Wood and paper products (export/import)	2002	1000 USD	40494	E	Estimated value
Import value	Wood and paper products (export/import)	2003	1000 USD	233876	E	Estimated value
Export value	Wood and paper products (export/import)	2003	1000 USD	32652	E	Estimated value
Import value	Wood and paper products (export/import)	2004	1000 USD	320050	E	Estimated value
Export value	Wood and paper products (export/import)	2004	1000 USD	52546	E	Estimated value
Import value	Wood and paper products (export/import)	2005	1000 USD	295802	E	Estimated value
Export value	Wood and paper products (export/import)	2005	1000 USD	54880	E	Estimated value
Import value	Wood and paper products (export/import)	2006	1000 USD	307959	E	Estimated value
Export value	Wood and paper products (export/import)	2006	1000 USD	54160	E	Estimated value
Import value	Wood and paper products (export/import)	2007	1000 USD	400033	E	Estimated value
Export value	Wood and paper products (export/import)	2007	1000 USD	94634	E	Estimated value
Import value	Wood and paper products (export/import)	2008	1000 USD	565721	E	Estimated value
Export value	Wood and paper products (export/import)	2008	1000 USD	73493	E	Estimated value
Import value	Wood and paper products (export/import)	2009	1000 USD	702153	E	Estimated value
Export value	Wood and paper products (export/import)	2009	1000 USD	101684	E	Estimated value

```
# Verify no NAs were created during conversion
sum(is.na(nigeria_wood_trade_clean$year))
```

```
[1] 0
```

5, 6 & 7. Arrange Columns and Filter

The analysis is only interested in Import and Export values for the `1000 USD` unit. Instead of reshaping the data, a simpler and more efficient approach is to filter the data to remove physical quantities, and select the exact columns needed.

```
nigeria_wood_trade_arranged <- nigeria_wood_trade_clean |>
  filter(unit == "1000 USD") |>
  select(
    year,
    flag_description,
    item,
    element,
    value
  )

head(nigeria_wood_trade_arranged, 7) |>
  knitr::kable()
```

year	flag_description	item	element	value
2000	Official figure	Wood and paper products (export/import)	Import value	181796
2000	Estimated value	Wood and paper products (export/import)	Export value	52276
2001	Official figure	Wood and paper products (export/import)	Import value	170423
2001	Estimated value	Wood and paper products (export/import)	Export value	36587
2002	Estimated value	Wood and paper products (export/import)	Import value	174679
2002	Estimated value	Wood and paper products (export/import)	Export value	40494
2003	Estimated value	Wood and paper products (export/import)	Import value	233876

8. Convert to Million Dollars

The current `value` column is in thousands of dollars (`1000 USD`). To convert this to millions, the value is divided by 1000 and stored in a new column called `value_million`.

```
trade_millions <- nigeria_wood_trade_arranged |>
  mutate(
    value_million = value / 1000
  )

head(trade_millions) |>
  knitr::kable()
```

year	flag_description	item	element	value	value_million
2000	Official figure	Wood and paper products (export/import)	Import value	181796	181.796
2000	Estimated value	Wood and paper products (export/import)	Export value	52276	52.276
2001	Official figure	Wood and paper products (export/import)	Import value	170423	170.423
2001	Estimated value	Wood and paper products (export/import)	Export value	36587	36.587
2002	Estimated value	Wood and paper products (export/import)	Import value	174679	174.679
2002	Estimated value	Wood and paper products (export/import)	Export value	40494	40.494

9. Summary Tables

To understand the dataset's composition and trade dynamics, five summary tables are produced.

9.1 Number of Observations by Trade Direction and Unit

This table shows the distribution of data points across the different trade directions and units.

```
observations_by_direction_unit <- nigeria_wood_trade_clean |>
  filter(element %in% c("Import value", "Export value", "Import quantity", "Export quantity"
  count(element, unit, name = "number_of_observations") |>
  arrange(element, unit)
```

```r
observations_by_direction_unit |>
  knitr::kable()
```

Table 2

element	unit	number_of_observations
Export quantity	m3	580
Export quantity	t	340
Export value	1000 USD	1345
Import quantity	m3	542
Import quantity	t	532
Import value	1000 USD	1502

9.2 Total and Average Value by Trade Direction and Year

This table aggregates the financial trade values (`1000 USD`) by trade direction and year, keeping units separate to avoid mathematical errors.

```r
trade_summary <- nigeria_wood_trade_clean |>
  filter(element %in% c("Import value", "Export value")) |>
  group_by(element, year, unit) |>
  summarise(
    total_value = sum(value, na.rm = TRUE),
    average_value = mean(value, na.rm = TRUE),
    number_of_observations = n(),
    .groups = "drop"
  ) |>
  arrange(year, unit, element)

head(trade_summary, 12) |>
  knitr::kable()
```

Table 3

element	year	unit	total_value	average_value	number_of_observations
Export value	2000	1000 USD	205345	4563.222	45
Import value	2000	1000 USD	872681	16465.679	53

Table 3

element	year	unit	total_value	average_value	number_of_observations
Export value	2001	1000 USD	143256	3183.467	45
Import value	2001	1000 USD	799525	14806.019	54
Export value	2002	1000 USD	151868	3374.844	45
Import value	2002	1000 USD	815555	15102.870	54
Export value	2003	1000 USD	125551	2790.022	45
Import value	2003	1000 USD	1075878	19923.667	54
Export value	2004	1000 USD	213623	4747.178	45
Import value	2004	1000 USD	1529042	28315.593	54
Export value	2005	1000 USD	227001	5044.467	45
Import value	2005	1000 USD	1455838	26959.963	54

9.3 Excluding Aggregates to Prevent Double-Counting

Before identifying the top five export and import products, it is crucial to remove hierarchical aggregate categories (e.g., "Sawnwood" which is the sum of "Sawnwood, coniferous" and "Sawnwood, non-coniferous").

Why Excluding Aggregates is Necessary

The FAOSTAT dataset is structured hierarchically, meaning it contains both leaf-level products (individual tradeable goods, like Sawnwood, non-coniferous) and aggregate categories (parent categories that sum up multiple products, like Sawnwood or Wood and paper products).

If a "Top 5 Products" analysis is run on the raw data without removing aggregates, it will produce mathematically misleading results due to double-counting. The aggregate categories will rise to the top of the ranking because their values simply equal the sum of the underlying leaf-level products. For example, listing both "Sawnwood" and "Sawnwood, non-coniferous" in a top 5 list is redundant.

To answer the question "which individual products have the largest trade value?", the aggregate parent categories must be excluded.

An explicit exclusion list is built based on the FAOSTAT item hierarchy.

```
aggregate_items <- c(
  # Top-level totals
  "Wood and paper products (export/import)",
  "Primary wood and paper products (export/import)",
  "Secondary processed wood products (export/import)",
  "Secondary processed paper products (export/import)",

  # Roundwood branch
  "Roundwood",
  "Wood fuel",
  "Wood fuel, all species (export/import, 1961-2016)",
  "Industrial roundwood",
  "Industrial roundwood, coniferous (export/import)",
  "Industrial roundwood, non-coniferous tropical (export/import)",
  "Industrial roundwood, non-coniferous non-tropical (export/import)",
  "Sawlogs and veneer logs",
  "Pulpwood, round and split (production)",
  "Other industrial roundwood",

  # Chips branch
  "Wood chips, particles and residues",

  # Sawnwood branch
  "Sawnwood",

  # Panels branch
  "Wood-based panels",
  "Fibreboard",

  # Pulp branch
  "Total fibre furnish",
  "Pulp for paper",
  "Wood pulp",
  "Chemical wood pulp",

  # Paper branch
  "Paper and paperboard",
  "Graphic papers",
```

```
  "Packaging paper and paperboard",

  # Secondary processed sawnwood
  "Further processed sawnwood (export/import)"
)
```

The explicit aggregate_items list above was created by reviewing the FAOSTAT item hierarchy and identifying three types of aggregate parent categories:

- Top-level totals: Items whose names imply the sum of the entire dataset (e.g., Primary wood and paper products).

- Semantic umbrellas: Items that group specific sub-categories but do not share the exact same name prefix (e.g., Wood-based panels contains Plywood and LVL, Particle board, and Fibreboard).

- Comma-separated parents: Items that serve as the parent to a comma-separated list (e.g., Sawnwood is the parent of Sawnwood, coniferous and Sawnwood, non-coniferous).

By filtering out these 26 aggregate items, the subsequent tables isolate the true, distinct tradeable products.

```
top_export_value <- trade_millions |>
  filter(
    element == "Export value",
    !item %in% aggregate_items
  ) |>
  group_by(item) |>
  summarise(
    total_export_million = sum(value_million, na.rm = TRUE),
    .groups = "drop"
  ) |>
  slice_max(order_by = total_export_million, n = 5, with_ties = FALSE) |>
  arrange(desc(total_export_million))

top_export_value |>
  knitr::kable()
```

Table 4

item	total_export_million
Industrial roundwood, non-coniferous	2155.495
Wood charcoal	947.438

Table 4

item	total_export_million
Roundwood, non-coniferous	931.726
Sawnwood, non-coniferous	387.792
Further processed sawnwood, coniferous (export/import)	121.730

9.5 Top 5 Import Products by Total Trade Value

Similarly, the five products with the largest total import value are identified.

```
top_import_value <- trade_millions |>
  filter(
    element == "Import value",
    !item %in% aggregate_items
  ) |>
  group_by(item) |>
  summarise(
    total_import_million = sum(value_million, na.rm = TRUE),
    .groups = "drop"
  ) |>
  slice_max(order_by = total_import_million, n = 5, with_ties = FALSE) |>
  arrange(desc(total_import_million))

top_import_value |>
  knitr::kable()
```

Table 5

item	total_import_million
Printing and writing papers	4993.444
Other paper and paperboard	3661.789
Wooden furniture (export/import)	1715.788
Case materials	1494.720
Plywood and LVL	1485.716

9.6 Number and Proportion of Observations by Flag Status

In the FAOSTAT database, every observation (a specific product, year, and trade direction) is assigned a "flag." This flag indicates the source and reliability of the data point. The three flags present in the Nigeria dataset are:

- A (Official figure): The data was reported directly by the Nigerian government or official customs authorities.

- E (Estimated value): FAOSTAT estimated the value because official data was missing, delayed, or incomplete.

- X (Figure from external organization): The data was sourced from a non-governmental organization, industry report, or international body.

The table below calculates the count (`n_observations`) and the percentage (`proportion`) of each flag category.

```
flag_summary <- nigeria_wood_trade_clean |>
  count(flag, flag_description, name = "n_observations") |>
  mutate(
    proportion = round(n_observations / sum(n_observations), 3),
    percent    = round(proportion * 100, 1)
  ) |>
  arrange(desc(n_observations))

flag_summary |>
  knitr::kable()
```

Table 6

flag	flag_description	n_observations	proportion	percent
X	Figure from external organization	2810	0.500	50.0
E	Estimated value	2402	0.428	42.8
I	Value imputed by a receiving agency	235	0.042	4.2
A	Official figure	169	0.030	3.0

Observation

The data quality assessment reveals a significant reliance on non-official sources. Exactly 50% of the observations are flagged as X (Figure from external organization), while 42.8% are E (Estimated). The remaining records are either imputed by a receiving agency (4.2% I) or represent official government figures (3.0% A).

Implication for Policy

Because only a small fraction (3%) of the data is officially reported by Nigeria, the absolute trade totals presented in this report should be treated as indicative estimates rather than exact figures. While the trends and rankings remain analytically valid as these international sources are generally good, they are secondary sources. They might miss informal cross-border trade,

miscalculate tariffs, or use different exchange rates than the Nigerian government.The policy team should cross-reference these FAOSTAT numbers with the Nigerian National Bureau of Statistics (NBS) before making binding fiscal or trade policy decisions.

The NBS is the official statistical agency of the Nigerian government. They collect data directly from the Nigeria Customs Service. Therefore, the NBS holds the "ground truth" for what actually crossed Nigeria's borders.

10. Total and Average Value by Trade Direction

To visualize the trends over time, line charts are used for both total and average values. Line charts are the standard choice for time-series data as they clearly highlight turning points, volatility, and long-term divergence.

year	element	total_value_million	average_value_million	number_of_observations
2000	Export value	205.345	4.563222	45
2000	Import value	872.681	16.465679	53
2001	Export value	143.256	3.183467	45
2001	Import value	799.525	14.806019	54
2002	Export value	151.868	3.374844	45
2002	Import value	815.555	15.102870	54
2003	Export value	125.551	2.790022	45
2003	Import value	1075.878	19.923667	54
2004	Export value	213.623	4.747178	45
2004	Import value	1529.042	28.315593	54
2005	Export value	227.001	5.044467	45
2005	Import value	1455.838	26.959963	54
2006	Export value	210.350	4.674444	45
2006	Import value	1507.717	27.920685	54
2007	Export value	387.658	8.614622	45
2007	Import value	1917.750	35.513889	54
2008	Export value	319.300	6.140385	52
2008	Import value	2688.327	44.070934	61
2009	Export value	401.218	7.715731	52
2009	Import value	3282.657	53.814049	61

Chart Choice

Line charts were selected because they are the standard visualization for time-series data, making it easy to track trends, turning points, and the divergence between imports and exports over the 26-year period. Two separate charts are used because the magnitude of the Total value is vastly different from the Average value; plotting them on the same axis would flatten the average line.

Total Trade Value Over Time

```
total_chart <- annual_trade_millions |>
  filter(element %in% c("Import value", "Export value")) |>
  ggplot(mapping = aes(x = year,
                       y = total_value_million,
                       colour = element)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 1.5) +
  scale_colour_manual(values = c("Import value" = "darkorange", "Export value" = "steelblue"
  labs(
    title = "Total Wood & Paper Trade Value by Direction (2000-2025)",
    subtitle = "Total value summed across all products per year",
    x = "Year",
    y = "Total value (million USD)",
    colour = "Trade Direction"
  ) +
  theme_bw()

total_chart
```

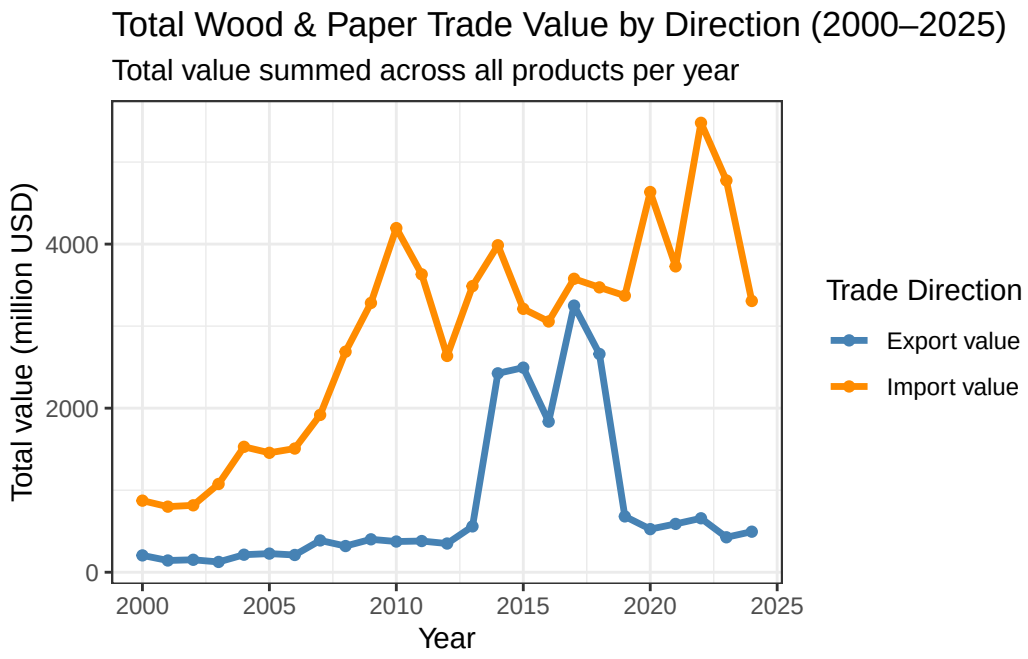


Figure 1: Total Wood & Paper Trade Value by Direction (2000–2025)

Average Trade Value Over Time

```
average_chart <- annual_trade_millions |>
  filter(element %in% c("Import value", "Export value")) |>
  ggplot(mapping = aes(x = year,
                       y = average_value_million,
                       colour = element)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 1.5) +
  scale_colour_manual(values = c("Import value" = "darkgreen", "Export value" = "darkred")) +
  labs(
    title = "Average Wood & Paper Trade Value by Direction (2000-2025)",
    subtitle = "Mean value per product-year record",
    x = "Year",
    y = "Average value (million USD)",
    colour = "Trade Direction"
  ) +
  theme_bw()

average_chart
```

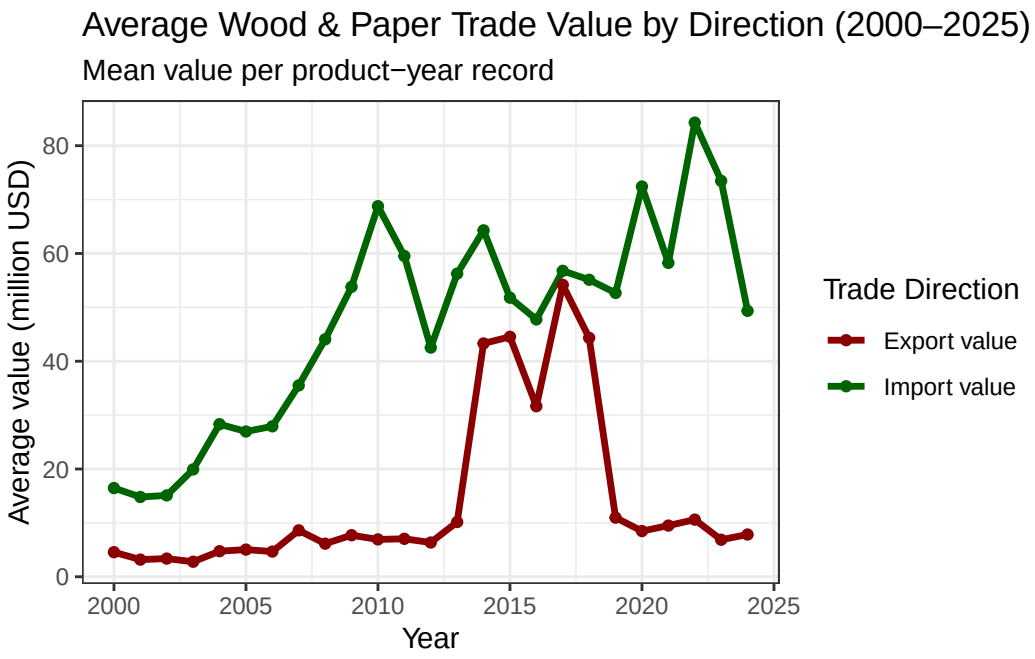


Figure 2: Average Wood & Paper Trade Value by Direction (2000–2025)

Observations

Total Value: Imports are much higher than exports every year, meaning Nigeria buys far more wood and paper products from other countries than it sells. The total trade value is very "uneven," with major spikes in import values around 2010 and 2021–2022. Export values remained low and stable until a sudden surge around 2014 and remain stable in 2015, decline afterward and experienced another surge in 2017 and experience a decline afterward.

Average Value: The average value looks almost identical to the total value, just on a much smaller scale (ranging from roughly 5 to 80 million USD). It goes up and down at the exact same times.

Why the trends differ

Over the 2000–2025 period, Nigeria started recording a much wider variety of smaller products in its trade data. Even though the Total money went up, the number of products being tracked went up at the exact same time.

However, the Average divides the total money by this growing list of products, adding many small, lower-value products dragged the mathematical average down.

Question 11: According to the total chart, export value was closest to import value during the periods of 2015 and 2017.

In both years, there were sharp surges in export value that significantly narrowed the trade deficit. The first spike occurred in 2015, and the second, even larger spike occurred in 2017, where the blue and red export line came closest to the orange and green import line.

12. Faceted Chart of Total and Average Values

Chart Choice

A faceted line chart was chosen using `facet_wrap()`. This allows the Total and Average charts to be displayed side-by-side while giving each its own independent Y-axis (`scales = "free_y"`). Without independent axes, the average line would appear completely flat because it is so much smaller than the total trade volume.

```
# 1. Reshape the data for faceting
facet_data <- annual_trade_millions |>
  filter(element %in% c("Import value", "Export value")) |>
  pivot_longer(
    cols = c(total_value_million, average_value_million),
    names_to = "measure",
    values_to = "value_million"
  ) |>
  mutate(
    measure = if_else(
```

```
      measure == "total_value_million",
      "Total value",
      "Average value"
    )
  )

# 2. Create the faceted chart
facet_chart <- ggplot(
  data = facet_data,
  mapping = aes(
    x = year,
    y = value_million,
    colour = element
  )
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 1.5) +
  facet_wrap(~ measure, scales = "free_y") +
  scale_colour_manual(values = c("Import value" = "darkgreen", "Export value" = "red")) +
  labs(
    title = "Total and Average Wood & Paper Trade Value\nby Direction (2000-2025)",
     subtitle = "Comparing total vs. average values (in million USD)\nby year and direction"
    x = "Year",
    y = "Value (million USD)",
    colour = "Trade Direction"
  ) +
  theme_light()

facet_chart
```

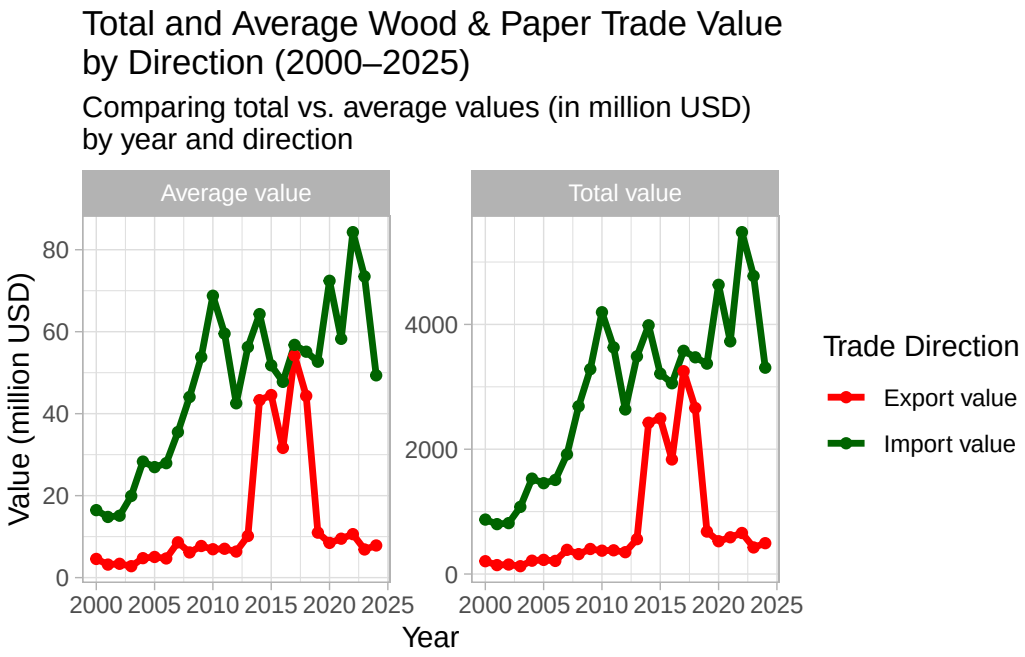


Figure 3: Faceted Comparison of Total and Average Trade Values

Observation

By placing the two charts side-by-side, it becomes immediately clear that the shapes of the two trends are almost identical, they peak and dip in the exact same years (e.g., the spikes around 2010, 2015, and 2022). However, the scale is drastically different. The Total value (right panel) operates in the thousands of millions, while the Average value (left panel) operates below 100 million.

13. Investigation into the Number of Products Recorded Over Time

Definition of a Counted Product: To investigate whether the number of recorded products changes over time, a "product" is defined as a unique item name (`item`) recorded in the dataset for a specific trade direction in a given year. The count is calculated using `n_distinct(item)`. This ensures that if the exact same product is recorded multiple times in a single year (e.g., due to different flags or units), it is only counted once.

```
# Calculate unique products per year and direction
product_counts <- nigeria_wood_trade_clean |>
  filter(element %in% c("Import value", "Export value")) |>
  group_by(year, element) |>
  summarise(
    unique_products = n_distinct(item),
```

```
    .groups = "drop"
  )

head(product_counts, 20) |>
  knitr::kable()
```

year	element	unique_products
2000	Export value	45
2000	Import value	53
2001	Export value	45
2001	Import value	54
2002	Export value	45
2002	Import value	54
2003	Export value	45
2003	Import value	54
2004	Export value	45
2004	Import value	54
2005	Export value	45
2005	Import value	54
2006	Export value	45
2006	Import value	54
2007	Export value	45
2007	Import value	54
2008	Export value	52
2008	Import value	61
2009	Export value	52
2009	Import value	61

```
# Create the line chart
product_counts_chart <- ggplot(
  data = product_counts,
  mapping = aes(x = year, y = unique_products, colour = element)
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 1.5) +
  scale_colour_manual(values = c("Import value" = "darkred", "Export value" = "navyblue")) +
  labs(
    # Added \n to prevent the title from cutting off in the PDF
    title = "Number of Unique Products Recorded\nby Year and Direction (2000-2025)",
    subtitle = "Tracking the expansion of product categories in FAOSTAT",
    x = "Year",
```

```
    y = "Number of Unique Products",
    colour = "Trade Direction"
  ) +
  theme_bw()

product_counts_chart
```

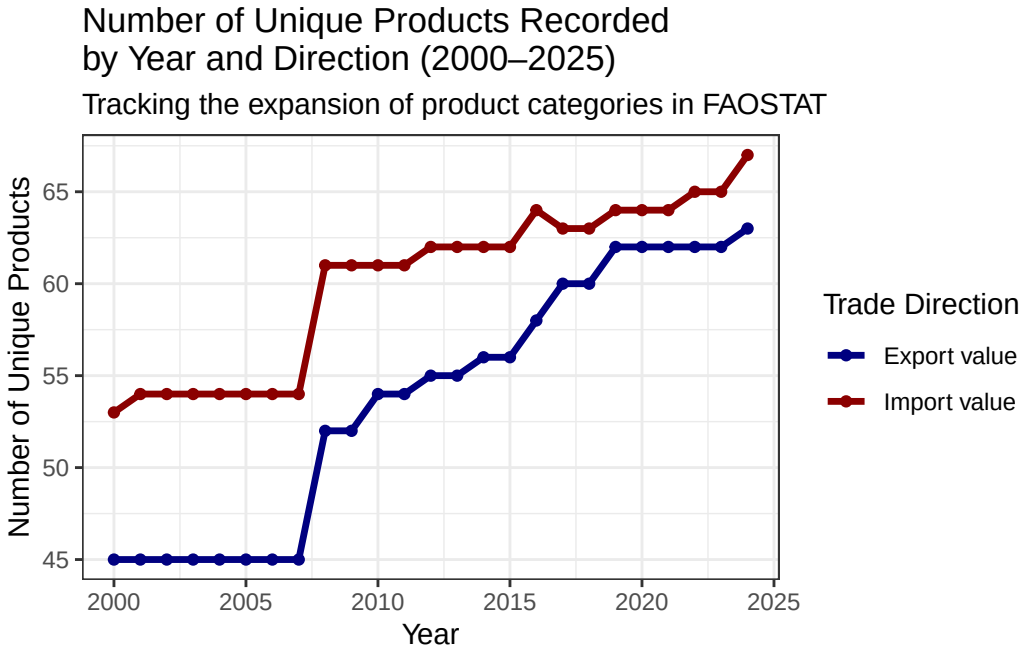


Figure 4: Number of Unique Products Recorded by Year and Direction

Observation

The number of products recorded does change significantly over time. The line chart displays a distinct "step-up" pattern. Between 2000 and 2007, the dataset tracked roughly 45 unique products for exports and 53 for imports. However, around 2008, the number of recorded products jumped sharply for both directions (reaching over 60). It continued to climb steadily through 2025, ending with over 65 unique products.

Implication

This confirms the composition effect discussed in previous sections. As FAOSTAT expanded its reporting coverage, more specialized and smaller product categories were added to the dataset. This increase in the denominator (the number of products) is the exact reason why

the average trade value per record remained relatively low, even while the total trade value in millions skyrocketed.

14. Plot of Top Five Export Products

A horizontal bar chart is used to visualize the top five export products. This chart type was chosen because product names in FAOSTAT are long and descriptive; a horizontal layout ensures the text remains perfectly readable without overlapping. Additionally, excluding aggregate items (as done in the data preparation) prevents double-counting, ensuring this chart reflects genuine, distinct products.

```
top_export_value |>
  mutate(item = fct_reorder(item, total_export_million)) |>
  ggplot(mapping = aes(x = total_export_million, y = item)) +
  geom_col(fill = "steelblue") +
  geom_text(
    mapping = aes(label = round(total_export_million, 1)),
    hjust = -0.2,
    size = 4,
    colour = "black",
    fontface = "bold" # Makes the numbers at the end of the bars bold
  ) +
  scale_x_continuous(expand = expansion(mult = c(0, 0.2))) +
  labs(
    title = "Top 5 Export Products by Total Trade Value\n(2000-2025)",
    subtitle = "Leaf-level products only (aggregates excluded)",
    x = "Total Export Value (Million USD)",
    y = NULL
  ) +
  theme_bw() +
  theme(
    # Bold Title and Subtitle
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(face = "bold", size = 11),

    # Bold Y-axis text (Product names)
    axis.text.y = element_text(face = "bold", colour = "black", size = 10),

    # Bold X-axis text (Numbers: 0, 500, 1000...)
    axis.text.x = element_text(face = "bold", colour = "black", size = 10),

    # Bold X-axis label ("Total Export Value")
    axis.title.x = element_text(face = "bold", size = 12),
```

```
  # Add margin to prevent cutting off
  plot.margin = margin(t = 10, r = 20, b = 10, l = 10)
)
```

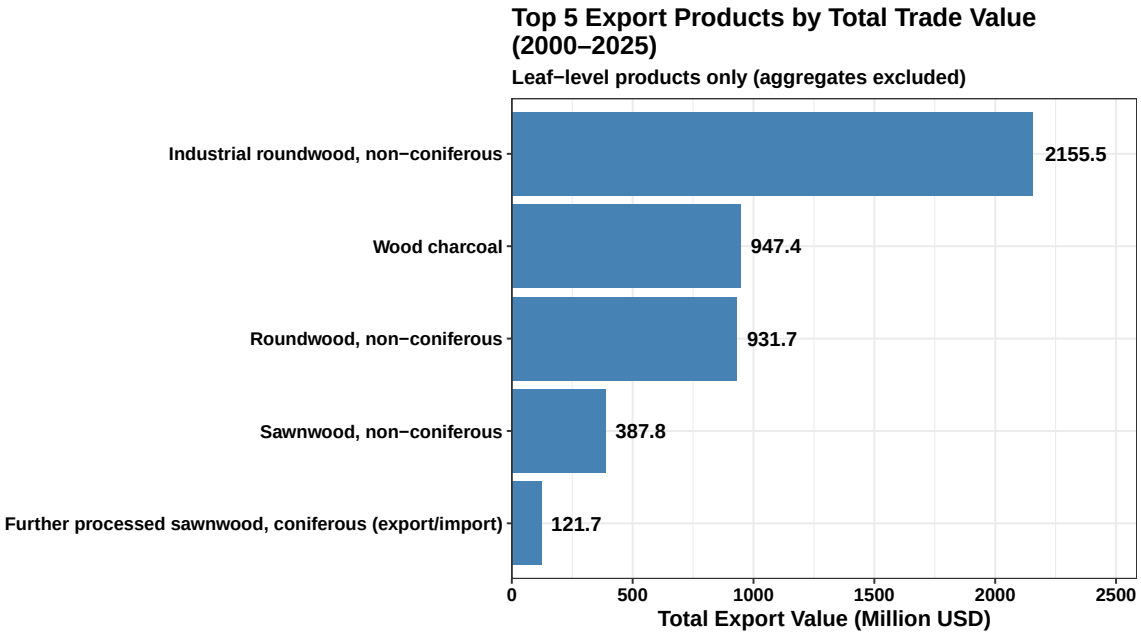


Figure 5: Top 5 Export Products by Total Trade Value (2000–2025)

15. Conclusion for a Non-Technical Policy Reader

This report analyzed Nigeria's wood and paper-product trade data from 2000 to 2025 (Food and Agriculture Organization, 2025). While the overall volume of trade has grown, the data reveals a persistent and widening trade deficit. Nigeria is importing far more finished goods than it is exporting, and its exports remain heavily reliant on raw, unprocessed materials.

Two Evidence-Based Findings

I. A Growing Trade Deficit: Imports consistently and significantly outpace exports. In recent years, import values have surged dramatically, peaking at 5,479 million USD in 2022, while export values remained a fraction of that total.

II. Export Concentration in Raw Materials: Nigeria's top exports are dominated by unprocessed wood. The largest export by far is Industrial roundwood (raw logs), followed by Wood charcoal. Meanwhile, higher-value finished products—like furniture or processed paper—make up a much smaller share of export revenue.

Recommendations

- Before implementing new tariffs or trade policies, the policy team should conduct a follow-up analysis with three specific objectives. First, because 97% of this dataset relies on external estimates, the team must cross-reference these FAOSTAT figures with official customs data from the Nigerian National Bureau of Statistics (National Bureau of Statistics, 2024) to validate the exact trade volumes. Second, the team should investigate the root causes of the massive import spikes (such as the surges in 2010 and 2022) to determine if they were driven by construction infrastructure projects (e.g., plywood/panels) or a lack of domestic paper manufacturing. Finally, the team should evaluate the return on investment for shifting from raw log exports to processed wood exports. Econometric studies show Nigeria's timber supply is depleting (Idumah et al., 2020); therefore, proving that exporting finished furniture yields higher revenue than raw Industrial roundwood would provide a strong economic justification for banning raw log exports and incentivizing local sawmills and processing plants.

References

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Idumah, F. O., Awe, F., & Orumwense, L. A. (2020). Dynamics of wood export in nigeria (1962-2017): An econometric analysis. *Russian Journal of Agricultural and Socio-Economic Sciences*, *97*(1), 212–218. https://doi.org/10.18551/rjoas.2020-01.24

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